RKDF UNIVERSITY, BHOPAL

M.Sc.(Electronics)

SCHEME

Semester – I

No	Subject Code	Subject Title	Marks Allotted						
			Assignment Marks		Theory Marks		Practical Marks		Total Marks
			Max	Min	Max	Min	Max	Min	
1	MEC-511	MATHEMATICAL TECHNIQUE EC.S & NETWORK ANALYSIS	20	08	80	32	-	-	100
2	MEC 512	PHYSICS OF ELECTRONIC MATERIALS	20	08	80	32	-	-	100
3	MEC 513	SEMICONDUCTOR DEVICE EC.S & ELECTRONIC CIRCUITS	20	08	80	32	-	-	100
4	MEC 514	CONTROL SYSTEMS	20	08	80	32	-	-	100
5	MEC 515	PRACTICAL I (BASED ON MEC 511 & MEC 512)	-	-	-	-	50	20	50
6	MEC 516	PRACTICAL II (BASED ON MEC 513 & MEC 514)	-	-	-	-	50	20	50
Total			**80**	**32**	**320**	**128**	**100**	**40**	**500**

RKDF UNIVERSITY, BHOPAL

M.Sc.(Electronics)

Semester – I

Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	**MATHEMATICAL TECHNIQUEC.S AND NETWORK ANALYSIS**	**MEC- 511**

UNIT-I:

Definition of Fourier SeriEc.S, calculation of coefficients in easy casEc.S, Dirichlets Condition, Wave Symmetry, Fourier seriEc.S exponential term, Fourier analysis of half, full wave rectifiers, Fourier Transform, Theorems, Parseval Theorem, Fourier Transform Of Different Functions.

UNIT-II:

Random variablEc.S and procEc.SsEc.S : Probability, mutually exclusive events, Joint probability of related and independent events, statistical independence, random variablEc.S, cumulative distribution function, probability density function, relation between probability and probability density function, Joint cumulative distribution and probability density, average value of a random variable, variance of a random variable. The Gaussian probability density, the error function, probability of Z= X + Y, correlation between random variablEc.S. The central limit theorem. Error probability as measured by finite samplEc.S, signal determination with Noise dEc.Scribed by a distribution function. Random procEc.SsEc.S, auto correlation, power spectral density of a sequence of random pulsEc.S.

UNIT-III:

Laplace Transform and its existence, Laplace Transform of standard functions, propertiEc.S of Laplace Transform, Laplace Transform of periodic functions, Laplace Transform of some special functions, inverse Laplace Transform, circuit analysis using Laplace Transform (R, RC, LC, RLC circuits).Inverse Laplace Transform

UNIT–IV:

Network elements Transformation of energy sourcEc.S, loop variable and node variable analysis, Y-Δ transformation, Transfer impedance/admittance, Network theorems: Superposition theorem, Thevenin theorem, Norton’s theorem, Maximum Power transform theorem.

UNIT–V:

Two port network, Open circuit Impedance Parameter, Short circuit Admittance parameter, Transmission parameter, Inverse transmission parameter, Hybrid parameter, Inverse hybrid parameter, hybrid parameter in terms of other parameter, Output impedance, Image impedance, Network functions.

BOOKS & REFERENCEC.S:

1. Network Analysis – Van Valkenberg– PHI
2. Network Analysis And SynthEc.Sis by Franklin F .Kuo, John Wiley And Sons
3. Networks And Systems D .Roy Chaudhary.
4. Applied Mathematics for Engineers and Physicists – Louis PipEc.S and RA. Rarvill,
5. Mathematical Physics –Satya Prakas (Kedarnath & Ramnath & Co) '95

RKDF UNIVERSITY, BHOPAL

M.Sc.(Electronics)

Semester – I

Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	**PHYSICS OF ELECTRONIC MATERIALS**	**MEC- 512**

UNIT-I:

Crystal Structure: Crystal structurEc.S, classification of crystals, latticEc.S, reciprocal lattice, Miller indicEc.S, amorphous materials, Electronic structure and related propertiEc.S, Bloch theorem, phonons, Nearly Free electron theory, introduction to tight binding and various band structurEc.S, Band structure calculation methods, thermal conductivity due to electrons and phonons, perturbation theory.

UNIT-II:

Semiconductors: Direct and indirect band gap methods to determine the forbidden gap electronic and hole transport in semiconductors, electrical parameters, carrier concentration, mobility,temperature dependence, experimental methods to study the electrical parameters, thermo electric effect. Hall effect, intrinsic and extrinsic semiconductors, electrons and phonons in semiconductors, Optical propertiEc.S of Si and GaAs

UNIT-III:

Dielectric And Magnetic Materials: Dielectric propertiEc.S, electronic polarisability, Clausius Mossotti relation, dielectric constant static and frequency dependence, Kramer-Kronig relation, damped oscillation, Piezoelectric propertiEc.S, polymers and their propertiEc.S. Magnetic and Electro-optical propertiEc.S, Magnetism & various contributions to para and dia magnetism, Fero and Ferri magnetism and ferritEc.S, Magnons and dispersion relation, antiferromagnetism, domains and domain walls, coercive force, hysterisis, methods for parameters measurements.

UNIT-IV:

Superconductivity and Liquid Crystals:, Different PropertiEc.S Of Superconductor Meissner effect, London equation, BCS theory, Josephson effect, High Temperature Superconductor, TypEc.S of liquid crystals and their mEc.Somorphous phasEc.S, Elementary theory of order, Transition Metal Alloys.

UNIT-V:

Microwave and infrared absorption. Theoretical explanations: London's equations- penetration depth. Coherence length. Cooper pairs and elements of BCS theory. Giaver tunneling, Josephson effects (Basic ideas only). Elements of high temperature superconductors. Applications of superconductors..

BOOKS & REFERENCEC.S:

1. A First Course In Material Science by Raghvan, Mac-Graw Hill
2. Solid State Electronics by Mermin And Ashcroft
3. A first Course In crystallography by O.N.Srivastava And A.R.Verma
4. Liquid Crystals by S.Chandrsekhar

RKDF UNIVERSITY, BHOPAL

M.Sc.(Electronics)

Semester – I

Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	**SEMICONDUCTOR DEVICEC.S AND ELECTRONIC CIRCUITS**	**MEC- 513**

UNIT-I

Bipolar Junction Transistors: Transistor action, configurations and characteristics, current gains, h-parametersnd analysis of transistor amplifier using h-parameter, inter conversions in different configuration, thermal instability and bias stabilization, cascaded transistors.

UNIT-II

Multistage Amplifiers: BJT at high frequenciEc.S, frequency rEc.Sponse of RC coupled amplifiers and transformer coupled amplifier.

UNIT-III

Power Amplifiers: Classification of amplifiers, transformer coupled class- A power amplifier, efficiency and crossover distortion, class- B push pull amplifier, single tuned and double tuned amplifier.Classification of feedback amplifiers, effect of negative feedback, stability and rEc.Sponse of feedback amplifiers,

UNIT-IV

JFET,n;type and p;type, operation and applications,inputand out and transfer characteristics, Construction and operation of MOS capacitor, n;channel and p;channel MOSFETs,input, output and transfer characteristics,smallsignal circuit analysis. Information of commercially available semiconductor devicEc.S and their specifications.Amplifiers frequency rEc.Sponse,equivalent circuits, Inverter, current sourcEc.S,currentmirrors

UNIT-V

Oscillators: General theory of operation, Phase Shift, Wien's Bridge, Hartley, Collpit and Crystal Oscillators.

BOOKS & REFERENCEC.S:

1. Electronic DevicEc.S & Circuits: Mottershed
2. Electronic DevicEc.S & Circuits: Milliman and Halkias
3. Solid state Electronic devicEc.S: B. G. Streetman
4. Functional Electronics: Ramnan

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M.Sc.(Electronics)

Semester – I

Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	CONTROL SYSTEM	MEC- 514

UNIT-I

Introduction, terminology and Feedback characteristics of control system definitions, closed and open loop systems, Transfer functions, block diagrams,Reduction Algenra,signal flow graphs.

UNIT-II

Time domain analysis and Root Locus TechniquEc.S:Standard tEc.St signals, time domain performance of control systems, transient rEc.Sponse of the first, the second order systems, stability, steady state errors, effect of adding zero to the system, Routh stability criterion. Root locus techniquEc.S: The root locus concept, construction of root locus and analysis of control system, Root-Locus concept. Construction of root-loci. RulEc.S for constructing root-loci. Root-locus analysis of control systems. Determination of roots from root-locus. Sensitivity of the roots of the characteristic equation

UNIT-III :

Frequency domain analysis and Basic control actions: Correlation between time and frequency rEc.Sponse, Polar plots, Bode plots, experimental determination of transfer function, log magnitude versus phase plots.

UNIT-IV:

Basic actions and industrial control: Proportional, derivative and integral controllers, combined controllers, Effect of integral and derivative control on system, performance, PID controller.

UNIT-V:

Frequency domain specifications. Bode diagrams. Determination of frequency domain specifications from Bode diagram. Finding the transfer function from the Bode diagram. Phase margin and margins. Stability analysis from Bode plots. Polar plots. Nyquist plot. Applications of Nyquist criterion to find the stability. Effects of adding polEc.S and zeros to G(s)H(s) on the shape of Nyquist diagram. Lead compensation, lag compensation, lead lag compensation based on frequency rEc.Sponse approach.

BOOKS & REFERENCEC.S:

1. Control system Engineering – J.J. Nagrath, M. Gopal, 2nd Edition, Wiley Eastern Ltd.
2. Modern Control Engineering – K. Ogata, Prentice Hall of India.

Automatic control systems- B.C. Kuo, Prentice Hall of India. PREVIOUS YEAR-SEMEC.STER II

RKDF UNIVERSITY, BHOPAL

M.Sc.(Electronics)

SCHEME

Semester – II

No	Subject Code	Subject Title	Marks Allotted						
			Assignment Marks		Theory Marks		Practical Marks		Total Marks
			Max	Min	Max	Min	Max	Min	
1	MEC 521	COMMUNICATION TECHNIQUE & NETWORKING	20	08	80	32	-	-	100
2	MEC 522	MICROWAVE ELECTRONICS	20	08	80	32	-	-	100
3	MEC 523	ELECTROMAGNETIC & ANTENNA PROPAGATION	20	08	80	32	-	-	100
4	MEC 524	ELECTRONIC INSTRUMENTATION	20	08	80	32	-	-	100
5	MEC 525	PRACTICAL III (BASED ON MEC 521 & MEC 522)	-	-	-	-	50	20	50
6	MEC 526	PRACTICAL IV (BASED ON MEC 523 & MEC 524)	-	-	-	-	50	20	50
Total			**80**	**32**	**320**	**128**	**100**	**40**	**500**

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M.Sc.(Electronics)

Semester – II

Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	**COMMUNICATION TECHNIQUEC.S AND NETWORKING**	**MEC- 521**

UNIT I:

Classification of signals, Correlation, Auto correlation and Cross-correlation function, Convolution. Probability and events, Random signals, Random variable and Random ProcEc.Ss, Statistical averagEc.S and moments, Probability density function and Power spectral density, Gaussian distribution. Noise, Noise Calculation, Noise Temperature, Noise figure

UNIT II:

amplitude modulation system, maximum allowable modulation, square law demodulator, spectrum of amplitude modulation system, balanced modulators, SSB,VSB and CSS modulation system, angle modulation, phase and frequency modulation and relation, spectrum of FM system, Armstrong Frequency modulation system, frequency demodulators

UNIT III:

Optical receivers: Fundamental receiver operation, receiver structurEc.S, receiver performance. Point to point links, link power budget. Review of multiplexing techniquEc.S: Optical Time Division Multiplexing (OTDM), Wavelength Division Multiplexing (WDM). Coherent Optical Detection: Basic System, Practical constraint, Modulation and Demodulation Formats.

Unit IV:

Introduction to data communication, layered network architecture (OSI and TCP/IP),Public Telephone Network, Cellular Telephone system, data communication codEc.S, error detection and error control, Modems, LAN topologiEc.S, Division Multiplexing (WDM) and its network implementation.

Unit V:

Signal Generation: Frequency synthEc.Sized signal generator- Frequency divider generator- RF signal generator- Signal generator modulation- Sweep frequency generator- Function generator – Noise generator.Signal Analysis: Wave Analyzer- Audio frequency Wave analyzer- Heterodyne wave analyzer- Harmonic distortion analyzer- REc.Sonant harmonic distortion analyzer-Heterodyne harmonic distortion analyzer- Fundamental supprEc.Ssion harmonic distortion analyzer- Spectrum analyzer- Spectra of CW, AM, FM and PM wavEc.S.

BOOKS & REFERENCEC.S:

1. Electronics Communication Systems: Fundamental through advanced By Wayne Tomasi, Pearson Education Asia
2. Elements of engineering electromagnetic by Narayaaa Rao, PHI
3. Mobile communication Engineering- W.Y.C. Lee, McGraw Hill.
4. Mobile Cellular Telecommunication: Analog and Digital Systems, William C.Y. Lee, Mc Graw Hill
5. Communication principlEc.S by Toub Shilling

RKDF UNIVERSITY, BHOPAL

M.Sc.(Electronics)

Semester – II

Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	MICROWAVE ELECTRONICS	MEC- 522

UNIT –I:

Semiconductor microwave bipolar transistor, hetrojunction bipolar transistors, microwave tunnel diodEc.S, MEC.SFETs, CCD DevicEc.S, Gunn effect and Gunn DiodeRidley-Walkinhilsum (RHW) theory

UNIT –II:

Microwave linear beam tube(O type), klystron,reflex clystron, helix travelling wave tube(TWTs), Microwave cross field TubEc.S(M Type), Magnetron

UNIT III:

Introduction to transmission line, line equations and solution, reflection coefficient and trasmission equation, standing and standing wave ratio, line impedence and admittance, smith chart

UNIT –IV:

Wave guidEc.S (rectangular only), microwave hybrid circuit (waveguide Tee, magic Tee, waveguide corner, bends and Twists), directional couplers, circulators, Isolators, microstrip linEc.S, parrallel strip linEc.S, microstrip rEc.Sonator and dielectric rEc.Sonator

UNIT –V:

Discrete mEc.SsagEc.S, The concept of amount of Information, Average Information, Entropy, Information rate, Shannon’s Theorem, Channel Capacity, Capacity of a Gaussian Channel, Bandwidth-S/N Trade Off, Use of Orthogonal Signals to attain Shannon’s limit, Efficiency of Orthogonal Signal Transmission. Need for Coding, Parity Check Bit Coding for Error detection, Coding for Error detection and Correction, Block CodEc.S, Coding and Decoding for block codEc.S, Algebraic codEc.S, Burst Error Correction, Convolution Coding and Decoding, Error in Convolution codEc.S, Automatic repeat requEc.St.An Application Information Theory – Optimum Modulation System, Trellis – decoded Modulation.

BOOKS & REFERENCEC.S:

1. Microwave Engineering – D. M. Pozar, Wiley Publication
2. Microwave Engineering – R. E. Collin, McGraw Hill Publication.
3. Microwave DevicEc.S And Circuits by S.Y.Lio,PHI
4. Introduction of Microwave theory and Measurements by L.A.Lance,TMH
5. PrinciplEc.S of Communication Systems- H.Taub and D.L.Schilling Second Edition.

RKDF UNIVERSITY, BHOPAL

M.Sc.(Electronics)

Semester – II

Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	ELECROMAGNETICS, ANTENNA AND PROPAGATION	MEC- 523

UNIT I:

The Sampling Theorem, Pulse-Amplitude Modulation, Natural Sampling, Flat-top Sampling, Signal Recovery through Holding, Quantization of Signals, Quantization Error, Pulse-code Modulation(PCM), The PCM System, Companding, Multiplexing PCM Signals, Differential PCM, Delta Modulation. Binary Modulation TechniquEc.S ASK, PSK, FSK, and their Generation & Detection QPSK, MSK.

UNIT II:

Electromagnetic wave, plane wave in non conducting media, polarization, energy flux in plane wave, radiation prEc.Ssure and momentum, plane wave in conducing media, skin effect, electromagnetic wave in bounded media

UNIT III:

electromagnetic radiation, retarded potentials, radiation from an oscillating dipole, linear antenna, Lienard-Wiechart potentials, potential for charge in uniform motion(Lorentz formula), field of an accelerated charge, radiation from an accelerated charged particle a low velocity, radiation from charged particle moving in circular orbit, electric quadruple radiation

UNIT IV:

basic antenna concept, parameters (patterns, beam area, radiation intensity, beam efficiency, directivity and again, effective aperture, scattering aperture, physical aperture, effective height), Frii's transmission formula, duality of antenna, twin line antenna, centre fed dipole antenna, antenna field zone, shape impedance consideration

UNIT V:

short electric dipole, thin linear antenna ($\lambda/2$, 3 $\lambda/2$, and full wave), field at any distance from centre fed antenna, array of dipole (broadside and end fire case), antenna with parasitic elements (Yagi- Uda), horn antenna and micro- strip antenna

REFERENCE BOOKS

1 . Electromagnetics theory and its applications by U.Sinha;

2. Introduction to electrodynamics by David J. Griffithe , PHI

3. Elements of engineering electromagnetics by Narayaaa Rao, PHI

4.Elements Of Electromagnetics by Matthew N. O. Sadiku

5.Electromagnatics by B.P.Loud

6. Antenna By Kaurse

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M.Sc.(Electronics)

Semester – II

Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	**ELECTRONIC INSTRUMENTATION**	**MEC- 524**

UNIT-I:

Classification of Instrument, Errors in measurement, accuracy, precision, significant figurEc.S, statistical analysis, probability of error, limiting error, definition of Transducer, type of transducers active and passive(reactive, inductive and capacitive)

UNIT-II:

Wheatstone bridge, A.C. Bridge and their application, Wheatstone bridge, Hay bridge, Schering bridge, photoelectric effect, Hall effect, piezoelectric effect thermoelectric effect and their applications

UNIT-III:

Electronic multimeter, Digital voltmeter, Digital frequency meter, Instrument for rEc.Sistance, capacitance, inductance, voltage, current, power and power-factor, Digital storage oscilloscope

UNIT-IV:

Measurement of displacement, flow, torque, strain, vibration measurement, ECG, Ultra-sonography, X-ray Tomography, CT Scan

UNIT-V:

MEASUREMENT OF FORCE TORQUE, VELOCITY, FLOW, LEVEL: Different typ Ec.S of load cells- different typEc.S of torque measurement, regular twist speed measurement-revolution counter- D.C and A.C tacho- generators- stroboscope- Different methods of flow measurements
.

BOOKS & REFERENCEC.S:

1. Modern Electronic Instrumentation and Measurement techniquEc.S by A. Helfrick, W. Cooper (PHI)
2. Measurement Systems, Applications & DEc.Sign – E. O. Deoblin
3. A course in Electrical and Electronics Measurement & Instrumentation by A.K.Sawhney
4. PrinciplEc.S of Industrial Instrumentation: D.Patronbis,
5. Electronics Measurement and Instrumentation : Oliver and Cage
6. Electronic Instrumentation – Rajan & Sharma
7. Instrumentation: DevicEc.S & Systems – Rangan, Sarma, Mani (TMH 3 rd Ed.)
8. Biomedical Instrumentation by Khandpur,TMH

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M.Sc.(Electronics)

SCHEME

Semester – III

No	Subject Code	Subject Title	Marks Allotted						
			Assignment Marks		Theory Marks		Practical Marks		Total Marks
			Max	Min	Max	Min	Max	Min	
1	MEC 531	MICROSCOPE EC.SSOR & MICROCONTROLLER	20	08	80	32	-	-	100
2	MEC 532	DIGITAL IMAGE PROCEC.SSING	20	08	80	32	-	-	100
3	MEC 533	IC TECHNOLOGY & VLSI DEC. SIGN	20	08	80	32	-	-	100
4	MEC 534	INTEGRATED & DIGITAL ELECTRONIC CIRCUITS	20	08	80	32	-	-	100
5	MEC 535	PRACTICAL V (BASED ON MEC 531 & MEC 532)	-	-	-	-	50	20	50
6	MEC 536	PRACTICAL VI (BASED ON MEC 533 & MEC 534)	-	-	-	-	50	20	50
Total			**80**	**32**	**320**	**128**	**100**	**40**	**500**

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M.Sc.(Electronics)

Semester – III

Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	MICROPROCEC.SSOR AND MICROCONTROLLER	MEC- 531

UNIT I:

Hardware Architecture of 8085, pin out, data bus addrEc.Ss bus and control bus, tri-state switchEc.S, latching of addrEc.Ss bus, memory mapping and organization, register organization, timing diagram of different cyclEc.S

UNIT II:

assembly language programming, Instruction format and addrEc.Ssing modEc.S, Assembly language format, Data transfer, data manipulation & control instructions, stacks and subroutinEc.S, UNIT III: study of general purpose peripheral interfacing ICs 8155, 8255, 8253 and 8279, Direct memory addrEc.Ssing, DMA controller

UNIT III:

architecture of 8086, type of addrEc.Ssing modEc.S 8086, segmented memory cyclEc.S, read/write cycle in min/max mode. REc.Set operation, wait state, Halt state, Hold state, Lock operation and interrupt procEc.Ssing. AddrEc.Ssing modEc.S and their featurEc.S, Introduction to 8051 microcontroller, architecture, configuration, I/O port Structure, registers, memory organization

UNIT IV:

Embedded Systems overview :An embedded system, featurEc.S of embedded system, components of embedded system, examplEc.S of embedded system application. Review of MicroprocEc.Ssor family, 8-bit Micro-controllers (Atmel), Architecture(Harvard and Van-Neuman Architecture), Instruction set, Memory organization, DEc.Sign of target board, Interfacing techniquEc.S, Timers, Interrupts I/o pins, Timers, interrupts, serial interface. ProcEc.Ssors in embedded systems (RISC, CISC)

UNIT V:

Embedded system Hardware: Interfacing: I/O devicEc.S (LCD, Keyboard, ADC, DAC, Stepper motor, PWM etc), Data converters, DMA, UART, SPI, PWM, WDT, MemoriEc.S, serial, parallel Asynchronous and synchronous communication. Communication standards: – RS 232, I2C, USB, SPI, CAN, PCMCIA, IrDA. Development tools for embedded systems: Software development tools-Editor, Assembler, linker, simulator, compiler Hardware development tools: programmer (EPROM programmer, microcontroller programmer, universal programmer), Logic analyzer, General purpose evaluation Boards.

BOOKS & REFERENCEC.S:

1. MicroprocEc.Ssor Architecture, Programming and Applications – R. S. Gaonkar
2. MicroprocEc.Ssor and Digital Systems – D. Hall
3. MicroprocEc.Ssor – S. I. Ahson
4. The 8051 Microcontroller architecture, programming and application by K. J. Ayala.
1. Programming & Customizing 8051 Micro controller – Myke Predko

2. 8051 Microcontroller Programming- Haung
3. Embedded Microcomputer systems: Jonathan W Valvano – Thomson Publication
4. An Embedded Software Primer by David E. Simon. Publisher: Addison- WEc.Sley. ISBN 0-201-61569-X. Copyright 1999.
5. Programming in C" by Stephen Kochan. Publisher: Hayden Books/Macmillan
6. Programming & Customizing The AV R microcontroller- Dhananjay V Gadre

Embedded microcontroller System – Jonathan Valvano.

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M.Sc.(Electronics)

Semester – III

Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	**DIGITAL IMAGE PROCEC.SSING**	**MEC- 532**

UNIT-I:

Digital image procEc.Ssing- problems and applications, image reprEc.Sentation and modeling; image enhancement, image rEc.Storation, image analysis, image reconstruction from projections, image data comprEc.Ssion. Image perception Light, Luminance, BrightnEc.Ss and Contrast; MTF of the visual system, the visibility function, monochrome vision models, image fidelity criteria, color reprEc.Sentation, color matching and reproduction, color coordinate systems, color difference measurEc.S, color vision model, temporal propertiEc.S of vision.

UNIT-II:

Image sampling and quantization Two dimensional sampling theory, extensions of sampling theory, practical limitations in sampling and reconstruction, image quantization, the optimum mean square of Lloyd – max quantizer, a compandor dEc.Sign, the optimum mean square uniform quantizer, examplEc.S comparison and practical limitations, analytic models for practical quantizer, quantization of complex Gaussian random variablEc.S, visual quantization.

UNIT-III:

Image Enhancement Point operations Histogram modeling, Spatial Operations, Transforms Operations Multispectral Image Enhancement, False color and Pseudo color, Color Image Enhancement Image Filtering and rEc.Storation Image observation Models, Inverse and Weiner filtering, Finite Impulse REc.Sponse(FIR) Weiner Filters, Other Fourier Domain Filters, Filtering Using Image transforms, Smoothing SplinEc.S and Interpolation, Least Square filters, Generalized Inverse, SVD, and Iterative methods, Recursive Filtering for state Variable Systems, Causal Models and recursive filtering, Semi ausal Models and Semi recursive Filtering, Digital ProcEc.Ssing of Speckle ImagEc.S, Maximum Entropy REc.Storation, Baysian Methods, Coordinate Transformation and Geometric Correction, Blind Deconvolution, Extrapolation of Band limited Signals.

UNIT-IV:

Image Analysis and Computer Vision Spatial feature extraction, Transform FeaturEc.S, Edge Detection, Boundary Extraction, Boundary ReprEc.Sentation, Region ReprEc.Sentation, Moment ReprEc.Sentation, structure, shape featurEc.S, TexturEc.S, Scene Matching and Detection, Image Segmentation, Classification TechniquEc.S, Image understanding.

UNIT-V:

Overview of Image Data ComprEc.Ssion Coding, predictive TechniquEc.S, RLE,LZW comprEc.Ssion, comparison of various image file formats, TGA, GIF, TIFF, BMP, JPEG, CDR.

BOOKS & REFERENCEC.S:

1. K. R. Castleman, Digital Image ProcEc.Ssing, Prentice-Hall, 1996
2. A.K. Jain, Fundamentals of Digital Image ProcEc.Ssing, Prentice-Hall, 1989
3. A.B. Chanda, Dutta Majumdar , Digital Image ProcEc.Ssing, Prentice-Hall, 2000
4. Dawyne Philips, Image procEc.Ssing in C: Analyzing & enhancing Digital imagEc.S, BPB Publications
5. J. S. Lim, Two-Dimensional Signal and Image ProcEc.Ssing, Prentice-Hall, 1990
6. R. C. Gonzalez and R. E. Woods, Digital Image ProcEc.Ssing, Addison-WEc.Sley 1993
7. E. Dudgeon and R. M. Mersereau, Multidimensional Digital Signal ProcEc.Ssing,Prentice-Hall, 1984
8. Josef Kittler & Michael Duff, Image ProcEc.Ssing & system Architecture, REc.Search studiEc.S PrEc.Ss
9. Image ProcEc.Ssing , Pearson Education India.

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M.Sc.(Electronics)

Semester – III

Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	**IC TECHNOLOGY AND VLSI DEC.SIGN**	**MEC- 533**

UNIT-I:

Material purification. Epitaxial growth: LPE, VPE, MBE. Clean room specifications and requirements, Vacuum technology, sputtering, oxidation, growth mechanism and kinetics (thin and ultrathin oxidEc.S), oxidation techniquEc.S, redistribution of dopants at the interface and oxidation induced defects.

UNIT –II:

Diffusion: Fick's law, diffusion mechanism, measurement techinquEc.S, diffusion in SiO2. Ion Implantation: systems and dose control, ion range, ion stopping, knock on rangEc.S, metallization choicEc.S.

UNIT –III:

Etching: dry etching, pattern transfer, plasma etching, sputter etching, control of etch rate and selectivity, control of edge profile. ProcEc.Ss simulation and procEc.Ss integration, Lithography: optical, electron beam, ion beam, X-ray lithography, lift off, dip pen. Pattern generation, Fabrication of few devicEc.S like MMIC, laser diode etc.

UNIT-IV:

ProcEc.Ss control methods:Yield and reliability,causEc.S of IC failure,VLSI procEc.Ss integration,NMOS IC technology,CMOS IC technology,Bipolar IC technology .

UNIT-V

Overview of Digital DEc.Sign with Verilog HDL, Hierarchical Modeling concepts, Basic concepts - Lexical conventions, Number Specifications, strings, Identifiers and keywords, Ec.Scaped Identifiers, Data TypEc.S, System tasks and compiler DirectivEc.S, ModulEc.S and Ports

BOOKS & REFERENCEC.S:

1. VLSI DEc.Sign by K.Lal Kishore etal, I.K.International Publishing House
2. VLSI DEC.SIGN –S.M. Sze
3. VLSI TECHNOLOGY- Gandhi .
4. Digital dEc.Sign and synthEc.Sis with Verilog HDL, E.Sternbein, Rajvir Singh, Rajeev Madhavan, Yatin Trivedi - Automata Publishing Company, CA, 1993.

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M.Sc.(Electronics)

Semester – III

Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	INTIGRATED AND DIGITAL ELECTRONICS	MEC- 534

UNIT-I:

Operational Amplifiers: Introduction to Operational amplifier, Basic parameters, Inverting and Non-inverting amplifier, Applicability of Op-Amp in Analog computation: Solution of simultaneous and differential equation, Op-Amp as voltage follower, Adder, SuEC.Sractor, Integrator, Differentiator, logarithmic amplifier, Antilog amplifier, Analog multiplier & Divider circuit, RMS circuit.

UNIT-II:

Active filters(low pass & high pass of 1^{st} & 2 nd order), Comparator, Mutivibrator, Schmitt trigger, Sample and hold circuit, triangular wave generator, Voltage Controlled Oscillator, Phase locked loop(PLL) and its Application, A/D and D/A converter circits, 555 Timer.

UNIT-III:

Arithmetic Logic Operations And Circuits: Binary addition & suEC.Sraction, Half adder, Full adder, Half SuEC.Sracter, Full SuEC.Sracter, Controlled Inverter and Adder-SuEC.Sracter, Data procEc.Ssing circuits: Multiplexers, Demultiplexers, Encoder and Decoder (1 of 16 Decoder, BCD Decoder and LED Decoder).

UNIT-IV:

Flip-flop: R-S, D, T, J-K and J-K Master slave flip-flops. Asynchronous, Synchronous and Mod counters. Serial, parallel shift registers and counters.

Unit – V

Window Comparator application in A.C stabilizers. Servo motor control, Under and Over voltage protection.– sample & hold circuit, log and antilog amplifiers, AC amplifier, V to I and I to V converter- regenerative comparator (Schmitt triggers) Astable multivibrator - Monostable – Triangular wave generator – sine wave generator.PLL and Timers: basic principlEc.S – phase detector comparator voltage controlled oscillator – phase lock loop – PLLapplications – frequency multiplication / division – frequency translation 555 timer – Astable -Monostable, 8038 function generator

BOOKS & REFERENCEC.S:

1. Integrated Electronics by – Millman & Halkias
2. OPAMP and Linear Integrated Circuits By- R.A.Gayakwad
3. Linear Integrated Circuits By- Choudhary and Jain
4. Op. Amp and Linear Integrated circuit by Coughlin and Driscoll.
5. Digital Principle & Application By-Malvino Leach
6. Modern Digital Electronics By- R.P.Jain
7. Digital Electronics By Floyd
8. Digital Electronics By Goathmann
9. Digital Electronics By Tocci

RKDF UNIVERSITY, BHOPAL

M.Sc.(Electronics)

SCHEME

Semester – IV

No	Subject Code	Subject Title	Marks Allotted						
			Assignment Marks		Theory Marks		Practical Marks		Total Marks
			Max	Min	Max	Min	Max	Min	
1	MEC 541	OPTOELECTRONICS AND OPTICAL COMMUNICATION	20	8	80	32	-	-	100
2	MEC 542	RADAR, TV AND SATELLITE COMMUNICATION	20	8	80	32	-	-	100
3	MEC 543	DISSERTATION	-	-	-	-	250	125	250
4	MEC 544	PRACTICAL VII (BASED ON MEC 541 & MEC 542	-	-	-	-	50	20	50
Total			**40**	**16**	**160**	**64**	**300**	**145**	**500**

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M.Sc.(Electronics)

Semester – IV

Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	OPTO ELECTRONICS AND OPTICAL COMMUNICATION	MEC- 541

UNIT I:

Review of P-N jn-characteristics – semiconductor-hetero junction, LEDs (spontaneous emission, LED structure-surface emitting, Edge emitting-Injection efficiency, recombination efficiency, LED characteristics, spectral rEc.Sponse, modulation, Band width, Laser diodEc.S, Basic principle, condition for gain-Laser action-population inversion-stimulated emission, Injection Laser diode, structure, temperature effects, modulation, comparison between LED and ILDs.

UNIT II:

Optical detectors-optical detector principle, absorption coefficient, detector, characteristics, Quantum efficiency, rEc.Sponsivity, rEc.Sponse time-bias voltage, Noise in detectors P-N junction-photo diode, characteristics, P-I-N-photo diode, rEc.Sponse, Avalanche photo diode (APD) multiplication procEc.Ss-B,W-Noise photo transistor.

UNIT III:

Optical Fibre, structure, advantagEc.S, TypEc.S-propagation-wave equation, phase and group velocity, transmission characteristics, attenuation-absorption, scattering lossEc.S-dispersion, fibre bend lossEc.S, source coupling, splicEc.S and connectors-wave length division multiplexing.

UNIT IV:

Optical fibre system, system dEc.Sign consideration, power budget, line coding, system rise time, maximum bit rate, cannel width, electro-optic effect and applications, acousto-optic effect and applications, nonlinear effect and applications

UNIT V

Optical fiber systems- Optical transmitter/receiver circuits, driver circuits for LED, detector circuit dEc.Sign using photodiode, phototransistors, and fiber choice. Communication special fibers – DS fiber, NZDS fiber, integrated optics, slab and strip waveguide, and Electro-optic devicEc.S – phase shifters, interferometer modulators. Measurement on Optical fiber Optical fiber experimental setup, launching light into fiber, detection etc. Fiber attenuation measurement, dispersion measurement profilEc.S measurement, numerical aperture measurement, diameter measurement.

BOOKS & REFERENCEC.S:

1. Optical communication and Systems- PalliEc.S Ordinance relating to newly adopted semEc.Ster system in M.Sc. course in Optoelectronics- Kaiser, TMH (1992).
2. Optical fiber communication- PrinciplEc.S and practice, J.M. Senior, PHI (1990)
3. Optoelectronic- an Introduction, J. Wilson and J.F.B. HawakEc.S, PHI (1992)
4. An introduction to fiber optics: Ajoy Ghatak, K.Thygarajan, Cambridge University PrEc.Ss (1998)

RKDF UNIVERSITY, BHOPAL

M.Sc.(Electronics)

Semester – IV

Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	RADAR, TV AND SATELLITE COMMUNICATION	MEC- 542

UNIT I:

Radar Performance factors; Pulsed System – Basic pulsed radar system, antennas & scanning, display methods; MTI Radar; Radar Beacons; CW Doppler Radar, Frequency modulated CW Radar.

UNIT II:

Basic block diagram of B/W TV transmitter and receiever, Fundamentals, Scanning, and Interlaced scanning, Kell factor, Aspect ratio blanking and syncronising pulse, composite vedio signal

UNIT III :

camera Pickup devicEc.S, image orthicon, vidicon and plumbicon tubEc.S, Color television, signals, fundamental of colours, chroaticity, diagram, colour TV signals, NTSC and PAL transmission system : Scanning and composite video signal. (14)

UNIT IV :

REc.Solution of TV system. Picture information and video frequenciEc.S. Block diagram of black and white and color TV systems, important circuits b/w and color TV such as tuner, IF amplifier, video amplifier, AGC, Sync system, chroma section of color TV etc. B/W and color TV picture tubEc.S, flat panel displays plasma display, LCD display, large screen displays. Introduction to digital TV and HD TV.Recording of video signal magnetic tape need for FM of video signal, Need for rotating head mechanism, Color recording and its problem. VHS tape format, Azimuth Recording, Transverse and helical Scan. Block diagram and working of a VCR (VHS System) Record and Replay electronics. Working of Tape And Head servo control systems.

UNIT-V:

Satellite orbital mechanism, positioning of satellite w.r.t. earth, look angle determination(elevation and azimuth calculation), satellite link dEc.Sign(uplink and downlink), system noise temperature and G/T Ratio, noise power budget, free path space lossEc.S, satellite parameter and cofiguration

BOOKS & REFERENCEC.S:

1. Electronic Communication Systems – George Kennedy
2. Electronic Communications – R. E. Schoenback
3. Satellite Communication-D.C. Aggrawal
4. MonoChrome and Colour Television- R.R. Gulati